Three-dimensional MHD evolution of an ultramassive white dwarf with differential rotation and a mixed field

Initial and final states at 192^3 and 256^3 — interim report

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Abstract

A $2.005\,M_\odot$ equilibrium with strong differential rotation and a toroidal-dominated field, evolved in three-dimensional ideal MHD at two resolutions. **The star survives and keeps its mass; its ordered field does not.** At 192^3, run to $t = 60\,\mathrm{s}$, the magnetic energy falls by a factor of 2170 and the toroidal-to-poloidal amplitude ratio from 338 to 1.07, while the mass stays within 0.3%. **The differential rotation is not erased**: both grids hold $\Omega_{\rm out}/\Omega_{\rm core}$ near 1/3 throughout, nowhere near the uniform rotation that would leave $2\,M_\odot$ unsupported, and at $t = 12\,\mathrm{s}$ they agree on it to 1.4%. The instability is $m = 1$ and grows in a few Alfvén times. Its rate is not converged — it rises 47% from 192^3 to 256^3 — and the post-saturation decay is not converged either, in the opposite direction; the finer grid holds twice the magnetic energy at $t = 12\,\mathrm{s}$. Both grids now reach $t \simeq 58\,\mathrm{s}$, and over that baseline the two central dynamical claims converge: **the profile steepens by 22.6% against 20.3%, and L_z brakes at half the rate once the field is gone — 0.200 against $0.214\%\,\mathrm{s}^{-1}$ while it lives, 0.087 against $0.094\%\,\mathrm{s}^{-1}$ after — so the identification of the transport as magnetic holds at two resolutions**. What does not converge is the leftover field: the residual magnetic energy is 25 times larger at 256^3 than at 192^3, and at 256^3 it is still slowly growing at $t = 58\,\mathrm{s}$.

1 What is being simulated

1.1 The star

1.2 The fields

1.3 Equation of state and method

Mass	M	$2.0052\,M_\odot$
	$M/M_{\rm Ch}$ ($\mu_e = 2$)	**1.38**
Central density	ρ_c	3.00×10^9 g cm^{-3}
Mean density	$\langle\rho\rangle$	3.30×10^7 g cm^{-3}
Equatorial radius	$R_{\rm eq}$	3.917×10^8 cm
Polar radius	$R_{\rm pol}$	1.501×10^8 cm
Oblateness	$R_{\rm pol}/R_{\rm eq}$	**0.383**
Volume-equivalent radius	$R_{\rm vol}$	3.07×10^8 cm
Rotation law	$\Omega(\varpi) = \Omega_c A^2/(A^2 + \varpi^2)$	$A = 1.834 \times 10^8$ cm
Central angular velocity	Ω_c	8.193 rad s^{-1} ($P = 0.77$ s)
Equatorial angular velocity	$\Omega(R_{\rm eq})$	1.47 rad s^{-1} ($P = 4.3$ s)
Degree of differential rotation	$\Omega_c/\Omega(R_{\rm eq})$	**5.6**
Equatorial velocity	$v_{\rm eq}$	6.2×10^3 km s^{-1} $= 0.021\,c$
Rotational support	$T/\lvert W\rvert$	0.0993
Mass shedding	$\Omega_{\rm eq}^2 R_{\rm eq}/g_{\rm eq}$	0.469
Dynamical time	$t_{\rm dyn}$	0.475 s
Alfvén time	t_A	2.383 s
Central rotation period	$P_{\rm rot}$	0.767 s

Table 1: The configuration. It lies 38% above the Chandrasekhar mass and can only exist because of the differential rotation, which is also why it is so flattened. It is not at breakup (47% of shedding) and is below both bar-mode thresholds ($T/|W| = 0.099$ against 0.14 secular and 0.27 dynamical). The surface period of 4.3 s is some three times shorter than the fastest white dwarf spin claimed in the literature; this is a merger-remnant-like configuration, not a model of an observed star.

	analytic model	192^3 grid	256^3 grid
$\max\lvert B_\varphi\rvert$ (G)	3.231×10^{13}	3.196×10^{13}	3.204×10^{13}
$\max\lvert B_{\rm pol}\rvert$ (G)	7.63×10^9	9.46×10^{10}	8.41×10^{10}
$\max\lvert B\rvert/B_c$	0.732	0.724	0.726
$E_{\rm mag}$ (erg)	6.035×10^{49}	6.035×10^{49}	6.056×10^{49}
$E_{\rm tor}/E_{\rm mag}$	0.99999995	0.999997	0.999998
$E_{\rm tor}/E_{\rm pol}$	$\mathbf{2.18 \times 10^7}$	3.11×10^5	5.22×10^5
$\max\lvert B_\varphi\rvert/\max\lvert B_{\rm pol}\rvert$	**4234**	338	381
$E_{\rm pol}/\lvert W\rvert$	5.2×10^{-10}	—	—
Toroidal filling factor	—	1.23%	—

Table 2: Mixed field, toroidal-dominated to one part in 10^7 by energy: a 10^9 G exterior dipole beside a 3×10^{13} G interior toroidal field. The peak toroidal field is reconstructed on the grid to 98.9% of the model value at 192^3; the poloidal component is not, because interpolating an axisymmetric model onto a Cartesian mesh injects poloidal field an order of magnitude above the true one. *The poloidal quantities at $t = 0$ are therefore numerical floors, not physics*, and the growth measured later is counted against them. The filling factor is $8\pi E_{\rm tor}/\max|B_\varphi|^2$ over the stellar volume.

Equation of state	`ztwd`: zero-temperature degenerate electrons, $\mu_e = 2$
	barotropic, $P = P(\rho)$; all $\partial/\partial T$ vanish, $c_V = c_P = 0$
Code	Castro 26.07 / AMReX; ideal MHD, constrained transport, HLLD
Grid	3D Cartesian, single level, box 1.8×10^9 cm per side
	192^3: d$x = 9.375 \times 10^6$ cm 256^3: 7.031×10^6 cm
Gravity	`PoissonGrav`, full 3D solve (the quadrupole is large at $R_{\rm pol}/R_{\rm eq} = 0.38$)
Initial condition	Grad–Shafranov/SCF on 385×769 (ϖ, z), interpolated from $\mathbf{A}$
CFL	0.5 at 192^3; 0.3 at 256^3 past $t \simeq 4.8$ s
Floor / sponge	$\rho_{\rm min} = 10^2$ against ambient 2×10^4; sponge 10^4–10^5
Diagnostics	rebuilt from B_x, B_y, B_z; the native `emag_density`,
	`etor_density` and `Div_B` derives are corrupt (see Sec. 7)

2 Results: the field is destroyed

The ordered field does not survive. It is disrupted by a non-axisymmetric instability within a few Alfvén times, and what is left after thirty seconds is a disordered residual three orders of magnitude weaker in energy. This section states what happens; how much of it depends on resolution is deferred to Sec. 6.

The field goes through three phases, all visible in Fig. 1.

Phase	Window	Rate	Signature
Growth	$t = 1.3$–$4.0\,\mathrm{s}$	$\sigma = 1.369\ \mathrm{s}^{-1}$ $(r = 0.994)$ e-fold of amplitude $0.61\,t_A$	E_{pol} exponential; $m = 1$ takes over
Saturation	$t \simeq 5.3\,\mathrm{s}$	—	E_{pol} peaks at $10^4\times$ its floor; $\max B_\varphi / \max B_{\mathrm{pol}} \to 1.19$
Decay	$t = 5.4$–$30\,\mathrm{s}$	e-fold 3.6–$4.0\,\mathrm{s}$	E_{tor} falls 10^3
Residual	$t > 30\,\mathrm{s}$	e-fold $37\,\mathrm{s}$	$E_{\mathrm{mag}} \simeq 3 \times 10^{46}$ erg, flat

Table 3: Phases of the field evolution at 192^3. The decay stalls after $t \simeq 30\,\mathrm{s}$: the field does not go to zero but settles at a disordered residual of order 10^{12} G.

The mode is $m = 1$. An azimuthal Fourier projection gives $|m{=}1|/|m{=}0| = 1.2 \times 10^{-16}$ at $t = 0$ — the initial condition is axisymmetric to machine precision. From $t \simeq 2.5\,\mathrm{s}$ the poloidal field is $m = 1$ dominated, with $|B_{z,m=1}|/|B_{z,m=0}|$ reaching 21 at $t = 4\,\mathrm{s}$, and the toroidal $m = 1$ saturates at $\simeq 10\%$ of the axisymmetric component. This rules out the axisymmetric MRI, which could not exist here in any case: built from the *vertical* field, $\lambda_{\mathrm{MRI}} = 2.9 \times 10^4$ cm, or 0.003 of a cell. What remains is the Tayler kink or the non-axisymmetric toroidal-field MRI, which the mode number alone cannot separate.

"Destroyed" means disordered, not weakened — until late. At $t = 5.2\,\mathrm{s}$ the peak toroidal field is *higher* than initially, 4.54×10^{13} against 3.20×10^{13} G, while the energy has fallen by a factor of three: the field is concentrating into filaments, its effective volume dropping from 1.49×10^{24} to $2.40 \times 10^{23}\ \mathrm{cm}^3$. Only afterwards does the peak collapse as well, to 1.23×10^{12} G.

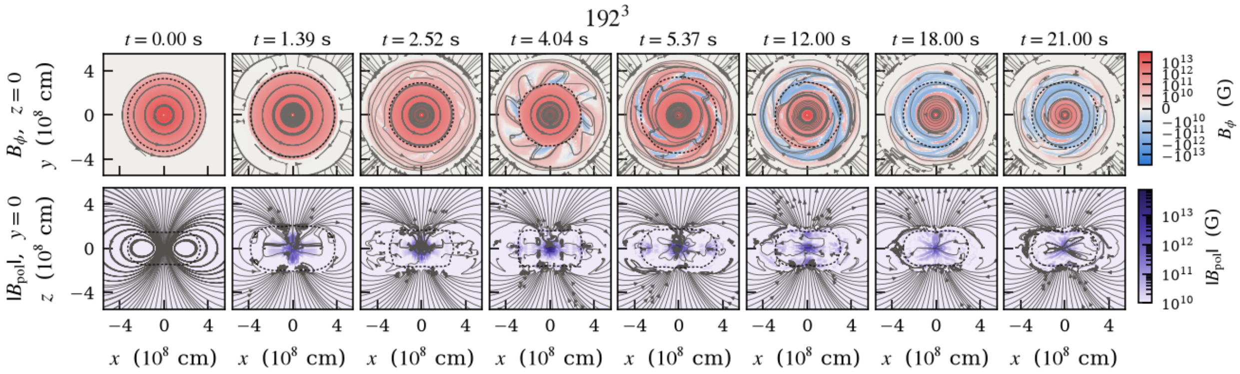

Figure 1: 192^3. **Top:** equatorial cut $z = 0$, where B_φ lies in the plane, so the streamlines *are* the toroidal field lines — circles while the star is axisymmetric. **Bottom:** meridional cut $y = 0$, where the in-plane field *is* the poloidal one. Colour is B_φ and $|B_{\rm pol}|$ respectively, both starting at 10^{10} G so the two rows are comparable; the near-empty poloidal panel at $t = 0$ is the statement that the poloidal field is three decades below the toroidal one. Same scales as Fig. 2.

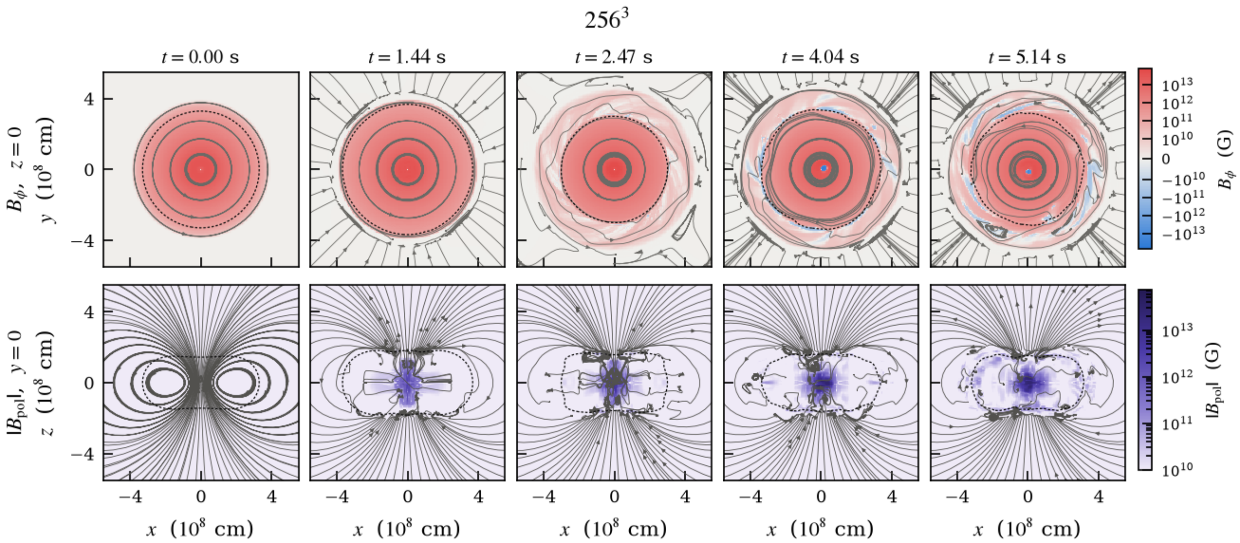

Figure 2: 256^3, same layout, planes and colour scales as Fig. 1.

3 Results: the star survives

The mass does not move. 0.02% from start to end (Table 4), 0.3% in amplitude along the way, while the magnetic energy of the same star falls by a factor of 2170. Whatever destroys the field does not unbind the star.

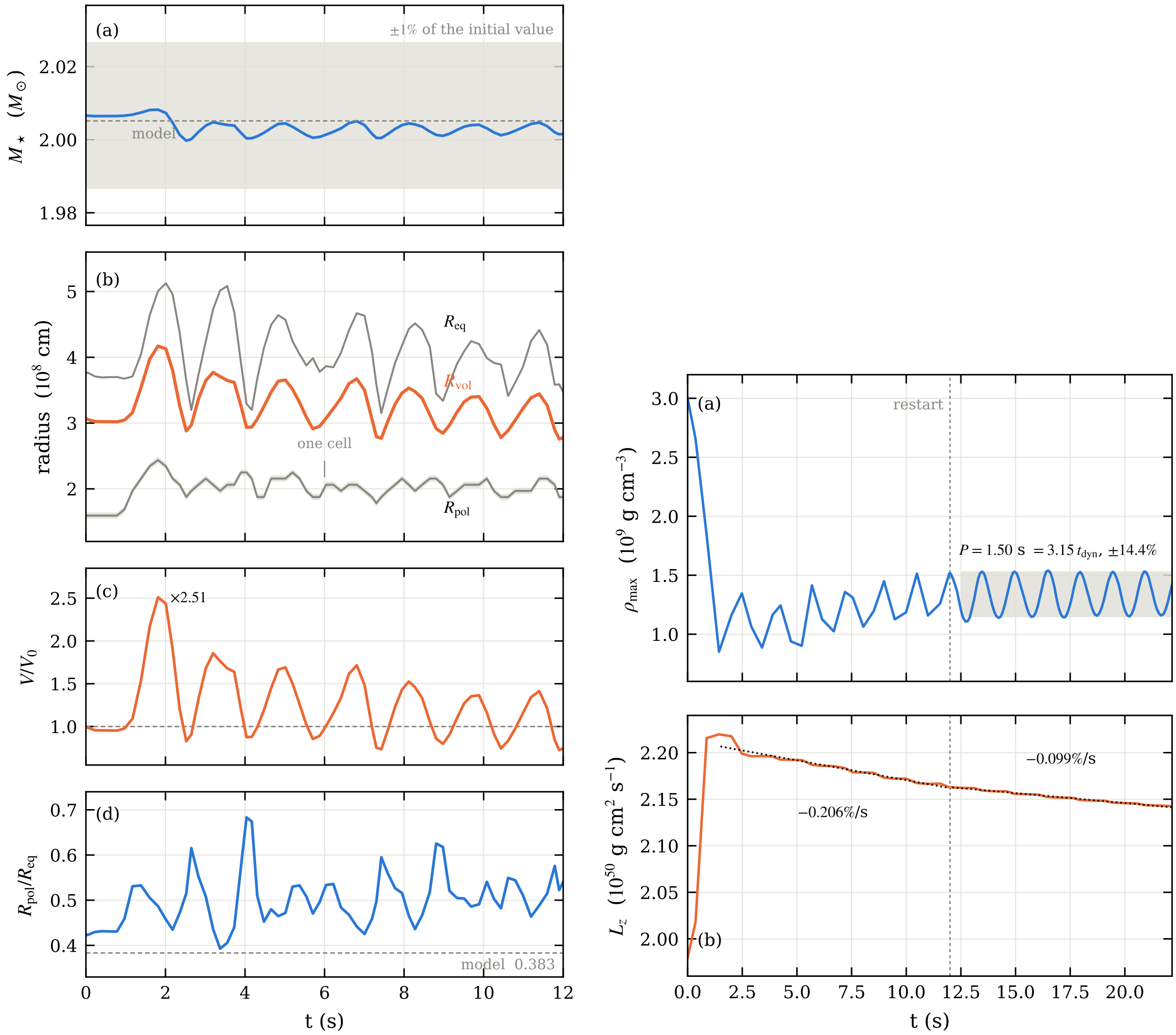

Figure 3: Left, 192^3 **to** $t = 12\,$**s:** mass against a ±1% band; the three radii with one grid cell drawn around $R_{\rm pol}$; volume; oblateness. **Right, to** $t = 22\,$**s:** central density, and axial angular momentum with the two braking rates fitted. The dashed vertical is the restart at $t = 12\,$s.

4 Results: rotation and angular momentum

This is the property the whole configuration depends on: a $2\,M_\odot$ white dwarf exists only because the rotation is differential, so what the field does to the rotation law decides whether the star has a future. Figures and numbers here are 192^3 over 60 s; Sec. 6.3 shows the 256^3 run doing the same thing, delayed and then faster, so the direction is a result and the magnitude is not yet.

The differential rotation not only survives, it strengthens. Fig. 4 is the diagnostic the long run was extended to provide. The magnetic transport is real and its signature is unambiguous: L_z falls at $0.21\%\,{\rm s}^{-1}$ while the toroidal field lives and $0.10\%\,{\rm s}^{-1}$ after it is gone. With no explicit viscosity and no explicit resistivity in these runs, nothing else in the scheme brakes anything, so that factor of two identifies the agent.

But the transport does not homogenise the star. Had it done so, $\Omega_{\rm out}/\Omega_{\rm core}$ would climb towards unity; it falls instead, from 0.34 to 0.27. The core holds its angular velocity to 3% over 60 s while the outer shell loses 25% of its own — angular momentum is carried outward and out of the star, and it is the outer layers that pay. Split by mass rather than by radius the same thing shows:

$L_z^{\mathrm{outer}}/L_z^{\mathrm{inner}}$ falls from 6.35 to 4.35, because the outer half of the mass loses 1% of its angular momentum while the star as a whole loses 5.5% and the inner half gains.

The star survives because the field never came close to removing what supports it, and in the time it existed it steepened the profile rather than flattening it. Both halves of that sentence hold at 256^3 as well (Sec. 6.3); what the finer grid does not reproduce is the size of the steepening over a fixed window, because it starts later.

Losing L_z and spinning up is expected here. $L_z = I\bar{\Omega}$ and I is not constant: between $t = 12$ and 60 s the star loses 2.8% of its angular momentum, contracts enough to lose 8.8% of its moment of inertia, and gains 6.6% in $\bar{\Omega}$ (Fig. 4b). This is not an oddity of the simulation. It is the behaviour known for rotating white dwarfs near the maximum mass, where a configuration shedding angular momentum contracts and spins up rather than spinning down — the spin-up branch discussed by Boshkayev et al. (2013). Their analysis is uniformly rotating and general relativistic while this configuration is differentially rotating, Newtonian and magnetised, so the correspondence is in the mechanism and not in the numbers.

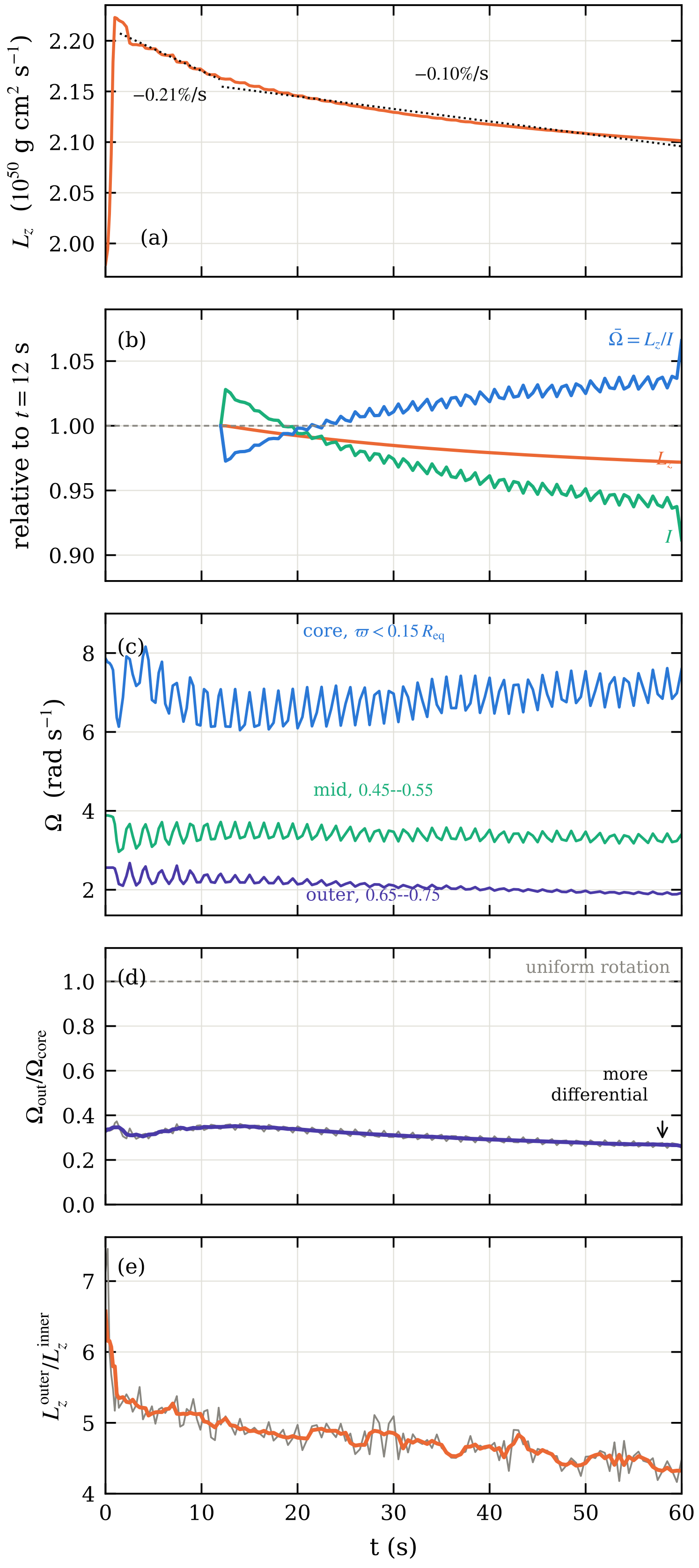

Figure 4: 192^3 over the full 60 s. **(a)** L_z of the star, with the two braking rates fitted; the 12% rise in the first second is the initial transient settling. **(b)** the decomposition $L_z = I\bar{\Omega}$, normalised at $t = 12$ s, which answers the obvious objection to (a): the star loses angular momentum and its angular velocity rises anyway, because I is not constant. Between $t = 12$ and 60 s, L_z falls 2.8%, I falls 8.8% as the star contracts, and $\bar{\Omega} = L_z/I$ gains 6.6% — a skater, minus the losses. **(c)** Ω mass-weighted in three cylindrical shells, all chosen to stay inside the star at every phase of the 1.5 s pulsation — note they sit at fixed fractions of the *initial* $R_{\rm eq}$, so part of the decline in the outer one is the star having shrunk out from under it. **(d)** the differential rotation; uniform rotation would be 1 and the star moves away from it. **(e)** L_z of the outer half of the *mass* over that of the inner half —7 split by mass rather than radius, with the dividing radius recomputed at every snapshot, because a shell at fixed ϖ would measure the pulsation instead of the transport. In (c) and (d) a running mean over one pulsation period is drawn over the raw curve; the ripple is the pulsation and aliases badly against coarse sampling.

5 The end state at 192^3

Only the 192^3 run has reached a steady late state; it was carried to $t = 60$ s and stopped changing well before that. The 256^3 run stands at $t = 40.5$ s and is continuing.

192^3	$t = 0$	$t = 38.5$ s	change
The star			
M ($M_\odot$)	2.00667	2.00618	**−0.02**%
$R_{\rm vol}$ (10^8 cm)	3.068	2.716	−11%
$R_{\rm eq}$ (10^8 cm)	3.750	3.204	−15%
$R_{\rm pol}/R_{\rm eq}$	0.425	0.556	rounder
$\Omega_{\rm core}$ (rad s^{-1})†	7.86	7.59	−3%
$\Omega_{\rm out}$ (rad s^{-1})†	2.56	1.92	−25%
$\Omega_{\rm out}/\Omega_{\rm core}{}^{\dagger\ddagger}$	0.341	0.266	−22%
L_z (10^{50} g cm^2 s^{-1})†	1.979	2.101	−5.5% from peak
$L_z^{\rm outer}/L_z^{\rm inner}{}^{\dagger\ddagger}$	6.35	4.35	−32%
I (10^{49} g cm^2)†	4.16	4.59	−8.8% from $t = 12$
The field			
$E_{\rm mag}$ (erg)	6.035×10^{49}	2.782×10^{46}	/ **2170**
$E_{\rm tor}$ (erg)	6.035×10^{49}	2.658×10^{46}	/ 2270
$E_{\rm pol}$ (erg)	1.942×10^{44}	1.236×10^{45}	×6.4
$E_{\rm tor}/E_{\rm pol}$	3.11×10^5	21.5	/ 1.4×10^4
$\max \lvert B_\varphi \rvert$ (G)	3.196×10^{13}	1.227×10^{12}	/ 26
$\max \lvert B_{\rm pol} \rvert$ (G)	9.46×10^{10}	1.148×10^{12}	×12
$\max \lvert B_\varphi \rvert / \max \lvert B_{\rm pol} \rvert$	**338**	**1.07**	/ 316
$\max \lvert B \rvert / B_c$	0.724	0.031	

Table 4: Initial and end state at 192^3. The star keeps its mass to 0.02%; the field is reduced by three orders of magnitude in energy and ends with its two components equal in peak strength. †Rotation and L_z are quoted at $t = 60$ s, the end of the run; the field and shape columns at $t = 38.5$ s, the last instant processed with the field diagnostics. ‡These two rows measure how the star redistributes angular momentum internally, and both converge over a long enough baseline: from $t = 12$ to 58 s the 256^3 run gives −20.3% and −6.2% against −22.6% and −10.6% here. Over shorter windows it appears not to, because its onset is some 6 s later (Sec. 6.3).

6 Resolution: what is measured and what is mesh

Everything above is physics as the two runs report it. This section asks which of it would survive a finer grid. The answer splits three ways, and the split is sharp enough to be useful: the *star* converges, the *energy* in the field does not, and the *rates* do not and move in opposite directions.

6.1 The same instant, and the same window

Table 5 takes the two grids at one instant, $t \simeq 5.2$ s, the end of the growth phase and the most violent moment of the run. It is the comparison that is easiest to make and the least trustworthy, and it is included to show why.

The 256^3 run has since reached its 12 s target, so the two grids can be compared over a window rather than at an instant. That matters: the star breathes by ±14% every 1.498 s, and any single snapshot catches an arbitrary phase of it. Table 6 averages over the last two periods, which removes the pulsation and leaves the state.

The rotation rows are the ones to take away. The conclusion that the star keeps its differential rotation — the property that lets it hold $2\,M_\odot$ — was previously measured at one resolution only. It is now measured at two, and they agree to 1.4%.

$t \simeq 5.2\,\mathrm{s}$	192^3	256^3	ratio
E_{tor} (erg)	1.965×10^{49}	1.784×10^{49}	0.91
E_{pol} (erg)	1.791×10^{48}	2.510×10^{48}	1.40
$E_{\mathrm{mag}}/E_{\mathrm{mag}}(0)$	35.5%	33.6%	
$E_{\mathrm{tor}}/E_{\mathrm{pol}}$	11.0	7.1	
$\max\lvert B_\varphi\rvert$ (G)	4.539×10^{13}	4.045×10^{13}	0.89
$\max\lvert B_{\mathrm{pol}}\rvert$ (G)	3.827×10^{13}	3.104×10^{13}	0.81
$\max\lvert B_\varphi\rvert/\max\lvert B_{\mathrm{pol}}\rvert$	1.19	1.30	
$\max\lvert B\rvert/B_c$	1.118	0.931	
M ($M_\odot$)	2.0035	2.0031	
R_{vol} (10^8 cm)	3.51	2.97	

Table 5: The two grids at the same physical time, at the end of the growth phase. They agree on the state to within tens of percent, but not on the rate that got there.

mean over 9–12 s	192^3	256^3	256/192
The star — converged			
M ($M_\odot$)	2.0028	2.0042	1.0007
R_{vol} (10^8 cm)	3.113	3.049	0.98
R_{eq} (10^8 cm)	3.895	3.883	**1.003**
$R_{\mathrm{pol}}/R_{\mathrm{eq}}$	0.514	0.501	0.98
V/V_0	1.06	0.99	0.93
Ω_{core} ($\mathrm{rad\,s^{-1}}$)	6.654	6.733	**1.012**
Ω_{out} ($\mathrm{rad\,s^{-1}}$)	2.294	2.357	**1.027**
$\Omega_{\mathrm{out}}/\Omega_{\mathrm{core}}$	0.345	0.350	**1.014**
L_z (10^{50} g cm^2 s^{-1})	2.168	2.078	0.96
$\max\lvert B\rvert/B_c$	0.771	0.826	1.07
Whole-star mass averages — less well converged			
$\bar{\Omega}$ ($\mathrm{rad\,s^{-1}}$)	3.992	4.360	1.09
$I = L_z/\bar{\Omega}$ (10^{49} g cm^2)	5.49	4.78	0.87
The field — not converged			
E_{tor} (erg)	4.07×10^{48}	8.00×10^{48}	**1.96**
E_{pol} (erg)	3.07×10^{47}	8.24×10^{47}	**2.69**
$E_{\mathrm{mag}}/E_{\mathrm{mag}}(0)$	7.3%	14.6%	2.01
$E_{\mathrm{tor}}/E_{\mathrm{pol}}$	13.3	9.7	0.73
$L_z^{\mathrm{outer}}/L_z^{\mathrm{inner}}$	5.04	5.90	1.17

Table 6: The two grids over the same three seconds, averaged over two pulsation periods. The split is sharp and it is the main new result of this report: everything about *the star* agrees to a few percent — mass, size, shape, and above all the rotation, where the core, the outer shell and their ratio all land within 3% — while everything about the *energy* in the field differs by a factor of two. The finer grid dissipates less, which is exactly what an ideal-MHD run with no explicit resistivity should do. The middle block is the honest exception: $\bar{\Omega}$ is mass-weighted over the whole star, so it is set partly by the tenuous outer layers just above the density cut, which are the least resolved part of the configuration. It converges to 9% rather than 1–3%, and $I = L_z/\bar{\Omega}$ inherits that. The shell measurements are local and do not.

6.2 The verdicts

One row deserves reading twice. The amplitude ratio $\max B_\varphi/\max B_{\mathrm{pol}}$ dips below unity only at 256^3 — the poloidal field overtakes the toroidal in peak strength there and never does at 192^3 — and at $t = 12\,\mathrm{s}$ it stands at 1.06 on the fine grid against 2.93 on the coarse. The 192^3 run does not reach 1.07 until $t = 38.5\,\mathrm{s}$ (Table 4). In terms of field topology the fine grid at 12 s is where the coarse grid is at nearly 40 s: refinement does not merely sharpen the picture, it advances the evolution. That is one more reason to treat every rate quoted here as a bound.

There is no explicit resistivity in these runs, so the decay of E_{tor} *is* the numerical resistivity,

	192^3	256^3	verdict
Mass range ($M_\odot$)	1.9997–2.0082	2.0021–2.0073	converged
$R_{\rm vol}(t=0)$ (10^8 cm)	3.068	3.069	converged (0.03%)
$\Omega_{\rm core}$, 9–12 s (rad s^{-1})	6.654	6.733	converged (1.2%)
$\Omega_{\rm out}/\Omega_{\rm core}$, 9–12 s	0.345	0.350	**converged (1.4%)**
max $\lvert B \rvert$, 9–12 s (G)	3.40×10^{13}	3.65×10^{13}	converged (7%)
σ from $E_{\rm pol}$ (s^{-1})	1.369	2.007	+47%**, not converged**
σ from the $m=1$ mode (s^{-1})	1.968	2.752	+40%, same factor
$E_{\rm pol}(t=0)$, the floor (erg)	1.94×10^{44}	1.16×10^{44}	finer seed is smaller
min(max B_φ/ max $B_{\rm pol}$)	1.19	**0.805**	instability goes further
at $t=12$ s	2.93	**1.06**	fine grid $\sim 3\times$ further along
Decay rate γ of $E_{\rm tor}$, 6–12 s	0.282 s^{-1}	0.141 s^{-1}	/ 2.0**, not converged**
as an e-folding	3.5 s	7.1 s	
$E_{\rm mag}$ at $t=12$ s (erg)	4.4×10^{48}	8.8×10^{48}	$\times 2.0$
First volume peak	2.51 V_0	1.79 V_0	early pulsation is numerical

Table 7: Three verdicts. *Converged:* everything about the star — mass, size, peak field, and the whole rotation profile. *Not converged:* the growth rate rises with resolution while the decay rate falls, each by about a factor of two per refinement, so the measured growth is a lower bound and the measured decay an upper bound; the coarse grid exaggerates the violence at both ends. The early large-amplitude pulsation is largely the initial condition ringing on the mesh.

and the two grids measure it. Halving γ while dx falls by a factor 1.33 puts it at d$x^{2.4}$, consistent with the second-order scheme and extrapolating to zero. Read literally, that says the field decay seen here is a resolution artefact and the physical star would keep its field far longer — but a two-point fit cannot distinguish that from a decay that converges to something small but finite, and the growth rate moving the other way (d$x^{-1.3}$) warns that both ends of the evolution are still mesh-limited.

Settling the rate would need 384^3, about five times the cost of 256^3. The alternative, and probably the right one, is explicit resistivity, giving the problem a physical dissipation scale instead of letting the mesh set it; Castro does not currently implement it.

6.3 Past 12 s: three tests over a 46 s baseline

The 256^3 run reached $t = 58.2$ s, so the two grids now overlap over almost the whole evolution. Means of two pulsation periods at each end, so the ±14% breathing is averaged out rather than interpolated through.

12–15 s → 55–58.3 s	192^3	256^3	verdict
L_z	−2.6%	−2.9%	converged
$\Omega_{\rm out}$	−15.3%	−14.9%	converged (3%)
$\Omega_{\rm out}/\Omega_{\rm core}$	−22.6%	−20.3%	**converged (10%)**
$\Omega_{\rm core}$	+9.4%	+6.7%	same sign
$L_z^{\rm outer}/L_z^{\rm inner}$	−10.6%	−6.2%	same sign

Table 8: The steepening of the differential rotation, over 46 s. The amplitude that did not converge over shorter windows does converge here, to 10%: the 256^3 onset is late, not absent, and the lag washes out once the baseline is long enough to contain it. The outer-shell spin-down agrees to 3%.

The lag itself is visible only in segments, and it is why two earlier drafts of this report reached two different wrong conclusions — first that the steepening was a coarse-grid artefact, then that its amplitude was irreducibly resolution-dependent.

change in $\Omega_{\rm out}/\Omega_{\rm core}$	192^3	256^3
$t = 12 \to 18\,$s	-3.2%	$+\mathbf{2.1}\%$
$t = 18 \to 24\,$s	-5.1%	-3.0%
$t = 24 \to 30\,$s	-4.5%	$-\mathbf{5.6}\%$
$t = 30 \to 38\,$s	-3.8%	$-\mathbf{5.2}\%$
$t = 12 \to 26.5\,$s (a 14 s baseline)	-8.2%	-1.0%
$t = 12 \to 58.2\,$s (a 46 s baseline)	-22.6%	-20.3%

Table 9: The finer grid starts about 6 s later, then steepens faster. The last two rows are the same comparison made over two baselines: at 14 s the two grids appear to disagree by a factor of eight, at 46 s they agree to 10%. Neither number is wrong; the short one simply does not contain the onset. *A convergence test over a window shorter than the lag between the grids measures the lag, not the physics.*

The braking rate halves, on both grids. This is the strongest result of the comparison. The claim that the angular-momentum transport is magnetic rests on L_z falling faster while the toroidal field lives than after it is gone, and that was measured at one resolution. It is now measured at two, in three successive regimes:

$\mathrm{d}\ln L_z/\mathrm{d}t$		192^3	256^3	
$t = 1.5$–$11.9\,$s	(field alive)	-0.1996%/s	-0.2144%/s	agree to 7%
$t = 12$–$30\,$s	(field decaying)	-0.0873%/s	-0.0935%/s	agree to 7%
$t = 30$–$58\,$s	(residual)	-0.0446%/s	-0.0523%/s	agree to 15%

Table 10: Each regime brakes at about half the rate of the one before, and the two grids agree on every one of them. With no explicit viscosity and no explicit resistivity in the scheme, nothing else brakes anything — so the factor of two identifies the agent, and it does so at two resolutions.

The residual field does not converge, and it is not decaying. The one place where the longer baseline makes things worse. At 192^3 the magnetic energy settles at $\simeq 3 \times 10^{46}\,$erg after $t \simeq 30\,$s. At 256^3 it settles at $7 \times 10^{47}\,$erg — a factor of 25 higher — and the peak field at $\simeq 6 \times 10^{12}\,$G against $\simeq 10^{12}\,$G. Worse for interpretation, the finer grid's residual is not flat: fitted over $t > 45\,$s it *grows* at $+0.012\,\mathrm{s}^{-1}$, with $E_{\rm tor}$ and $E_{\rm pol}$ both rising.

This is consistent with the $\mathrm{d}x^{2.4}$ scaling of the decay: less numerical resistivity leaves more residual field. Taken at face value it says the "disordered residual" reported at 192^3 is a floor set by the mesh, not by the star, and that the true residual is larger — possibly much larger, since a run that is still gaining magnetic energy at $t = 58\,$s has not reached a final state at all. The obvious mechanism is available: the differential rotation survives and is *steepening*, so there is free shear energy to wind residual poloidal field back into toroidal. Whether that is a genuine dynamo or the last decade of a decay that the mesh cannot resolve is not something two resolutions can settle.

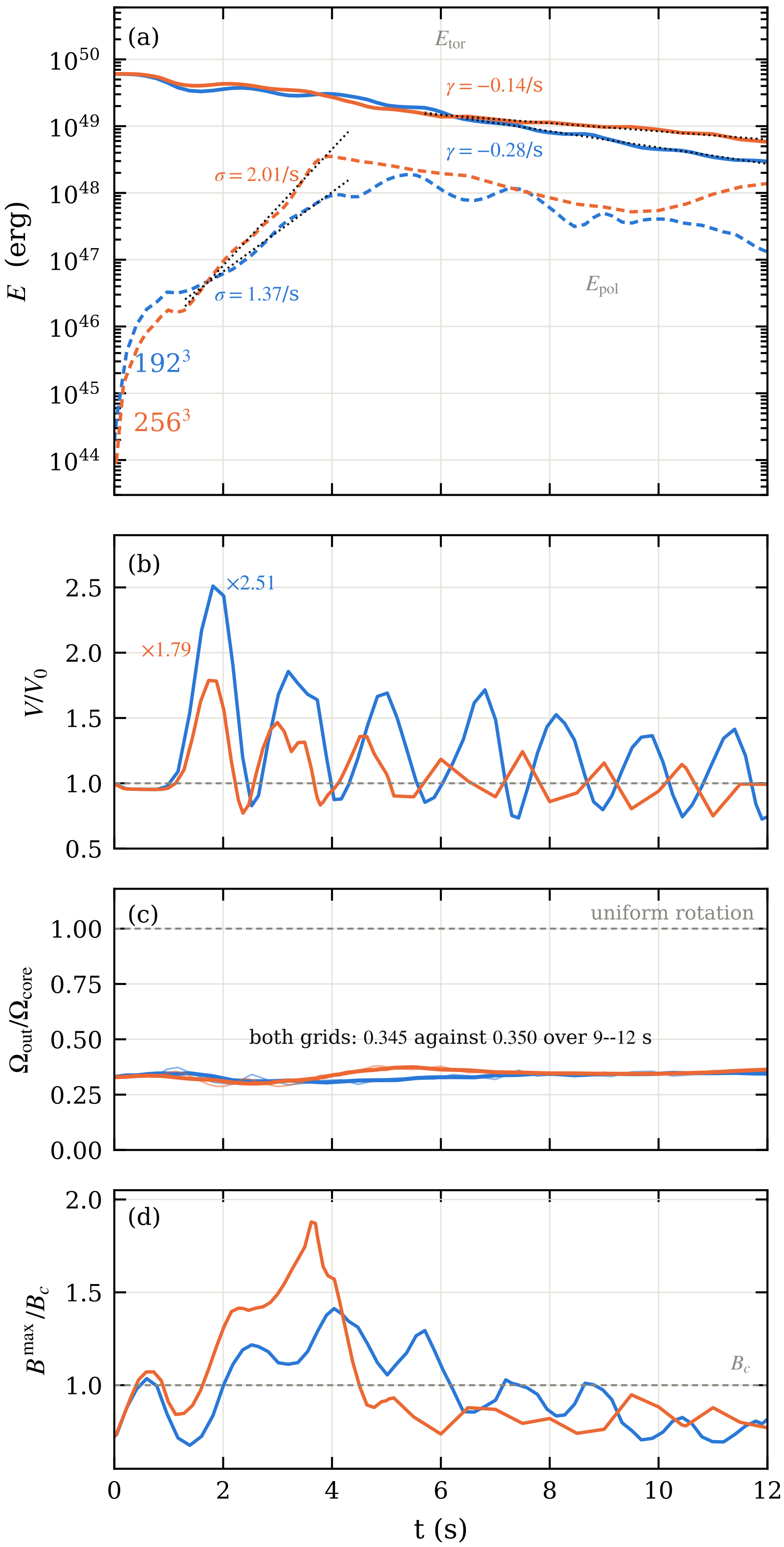

Figure 5: 192^3 against 256^3 over the full 12 s. **(a)** Toroidal (solid) and poloidal (dashed) energy, with the fitted growth σ and decay γ. The two rates move in *opposite* directions with refinement. **(b)** V/V_0: the first peak drops from 2.51 to 1.79, its excess over unity scaling as $dx^{2.3}$. **(c)** The differential rotation — neither grid approaches uniform rotation and the two agree to 1.4%. **(d)** Peak field against B_c; above the Landau field for roughly half the run at both resolutions.

7 Caveats

1. **The EOS is barotropic.** `ztwd` has no thermal reservoir, so the 6×10^{49} erg that leave the field have no physical destination and are dissipated at the grid scale. We can say the ordered field is destroyed; we cannot say where the energy went. The stratification is also neutral, with no buoyancy to limit the mode — in a stratified star the rate would be lower, making σ a *physical* upper bound while it is a *numerical* lower bound.

2. **Part of the run is outside the EOS validity range.** $\max|B|$ reaches $1.41\,B_c$ at $t = 4.0\,\mathrm{s}$ at 192^3 and $1.88\,B_c$ at $t = 3.6\,\mathrm{s}$ at 256^3, exceeding the Landau critical field in 44% and 48% of the sampled instants respectively. This does *not* improve with resolution — the finer grid goes further above B_c, not less far. The late state, at $0.03\,B_c$, is well inside.

3. **The poloidal field at $t = 0$ is a numerical floor.** Cartesian interpolation of an axisymmetric model injects E_{pol} some 70 times above the model's value even at 256^3. All poloidal growth is measured against that floor, which is why it is quoted alongside.

4. **A short baseline can invert the sign of a delayed effect.** The 256^3 steepening begins about 6 s after the 192^3 one. Measured to $t = 26.5\,\mathrm{s}$ the two grids appear to disagree by a factor of eight; measured to $t = 58\,\mathrm{s}$ they agree to 10%. Two earlier drafts of this report drew two different wrong conclusions from the short window (Sec. 6.3). Any convergence test here needs a baseline longer than the lag between the grids, and the lag is not known in advance.

5. **The residual field is not converged and is not settled.** At 192^3 the magnetic energy plateaus at $\simeq 3 \times 10^{46}$ erg; at 256^3 at 7×10^{47} erg, twenty-five times higher, and still *rising* at $t = 58\,\mathrm{s}$. Everything this report says about a "disordered residual" therefore describes a mesh-set floor rather than a state of the star, and the 192^3 factor of 2170 in E_{mag} is an upper bound on the destruction. Whether the finer grid is showing a genuine shear dynamo — the differential rotation is still there and still steepening, so the free energy exists — or merely the last decade of a decay it cannot resolve, two resolutions cannot say.

6. **Native magnetic diagnostics are unusable.** `emag_density`, `etor_density` and `Div_B` pack face-centred components of different index types into one cell-centred FAB and read past the end of the box, `Div_B` reaching 3×10^{146} where it should be ~ 0. Everything here is rebuilt from B_x, B_y, B_z, which are correct.

Status

192^3: complete, $t = 60\,\mathrm{s}$. 256^3: $t = 58.2\,\mathrm{s}$, fully processed. A final window to 60 s is queued; nothing here depends on it.

Inputs, submission scripts, diagnostic tools, data and figure code are under version control in the `WD_MAG` repository.

References

Boshkayev, K., Rueda, J. A., Ruffini, R., & Siutsou, I. 2013, *On General Relativistic Uniformly Rotating White Dwarfs*, ApJ, 762, 117.